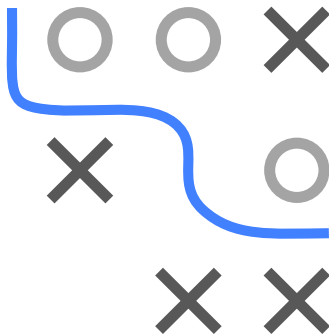




Introduction to Machine Learning: Nutshell

All slides

May 22, 2026

Introduction to Machine Learning

ML Basics

Supervised Regression

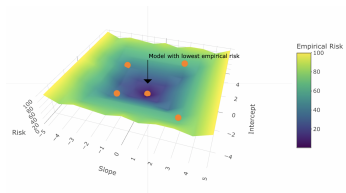
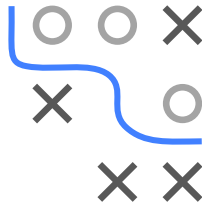
Supervised Classification

Random Forests

Neural Networks

Introduction to Machine Learning

ML-Basics In a Nutshell



Learning goals

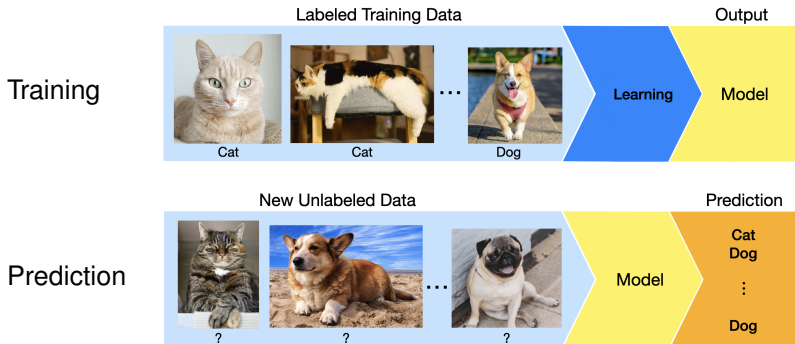
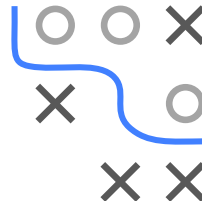
- Understand fundamental goal of supervised machine learning
- Know concepts of data, task, model, parameter, learner, loss function, and empirical risk minimization

WHAT IS ML?

“A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .”

Tom Mitchell, Carnegie Mellon University, 1998

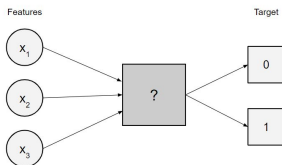
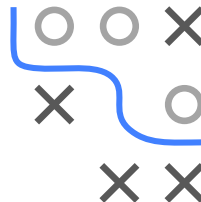
⇒ 99 % of this lecture is about **supervised learning**:



DATA IN SL

Measurements on different aspects of n objects:

- **Target:** output variable / goal of prediction
- **Features:** properties that describe an object



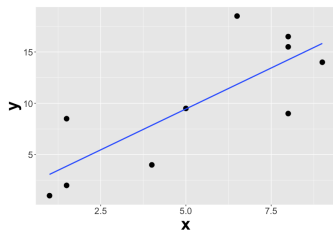
Features $\mathbf{x}$				Target y
Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
4.3	3.0	1.1	0.1	setosa
5.0	3.3	1.4	0.2	setosa
7.7	3.8	6.7	2.2	virginica
5.5	2.5	4.0	1.3	versicolor

$$\mathcal{D} = \left(\left(\mathbf{x}^{(1)}, y^{(1)} \right), \dots, \left(\mathbf{x}^{(n)}, y^{(n)} \right) \right) \in (\mathcal{X} \times \mathcal{Y})^n$$

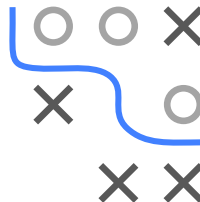
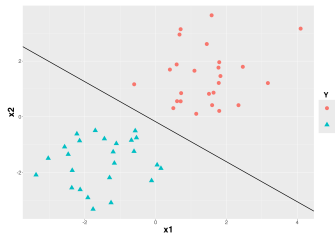
TASKS

- Supervised tasks are labeled data situations with the goal of learning the functional relationship between features and target
- We distinguish between **regression** and **classification** tasks, depending on whether the target is **numerical** or **categorical**

Regression: Target is **numerical**, e.g., predict days a patient has to stay in hospital



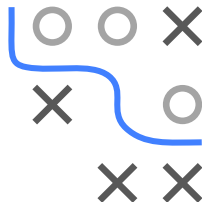
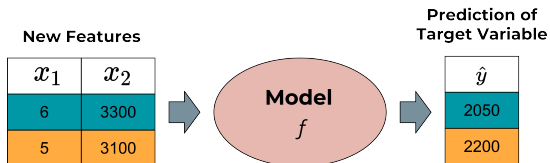
Classification: Target is **categorical**, e.g., predict one of two risk categories for a life insurance customer



MODELS AND PARAMETERS

- Maps features to predicted targets

$$f_{\theta} : \mathcal{X} \rightarrow \mathbb{R}^g$$



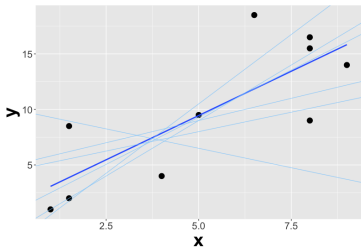
- We consider a restricted set of parametrized functions of a certain form, the **hypothesis space**, e.g., simple linear functions
- Models are fully determined by parameters. E.g., in the case of linear functions, $y = \theta_0 + \theta_1 x$, the parameters θ_0 (intercept) and θ_1 (slope) determine the relationship between y and x
- Finding the optimal model means finding the optimal set of parameters

LEARNER

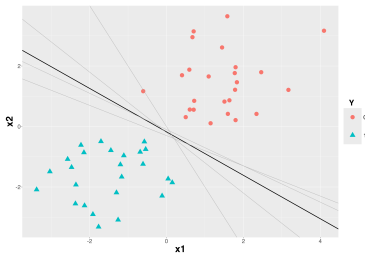
-

$$\mathcal{I} : \mathbb{D} \times \Lambda \rightarrow \mathcal{H}$$

Regression

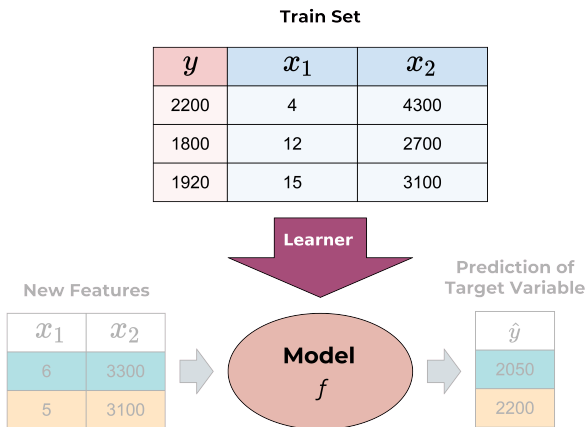
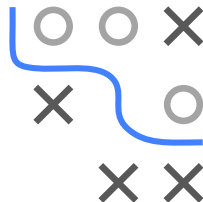


Classification



LEARNER

- Learner uses labeled training data to learn a model f . This model is applied to new unlabeled data for predicting the target variable



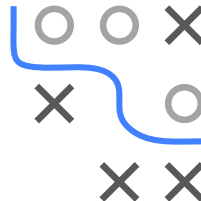
LOSS AND RISK MINIMIZATION

- Loss: Measured pointwise for each observation, e.g., L_2 -loss

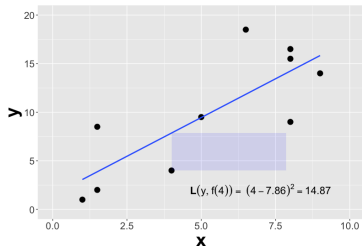
$$L(y, f(\mathbf{x})) = (y - f(\mathbf{x}))^2$$

- Risk: Measured for entire model. Sums up pointwise losses

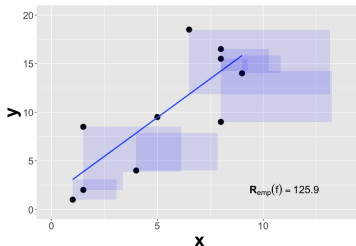
$$\mathcal{R}_{\text{emp}}(f) = \sum_{i=1}^n L(y^{(i)}, f(\mathbf{x}^{(i)}))$$



Squared **loss** of one **observation**



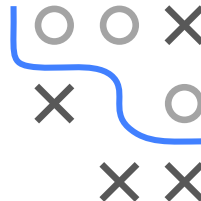
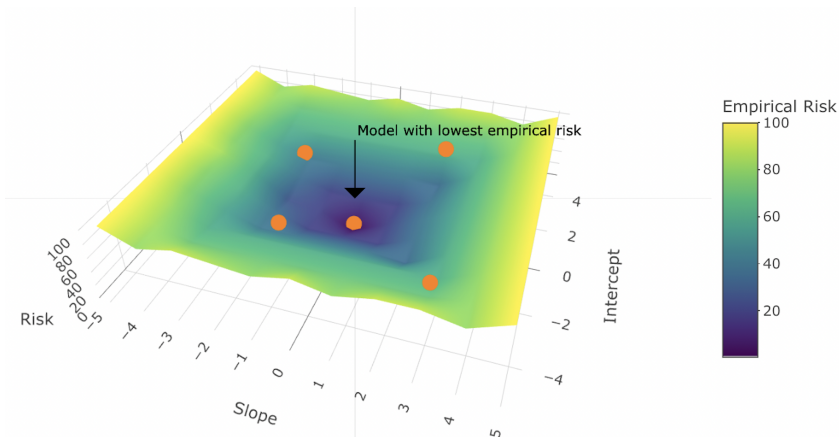
Empirical **risk** of entire **model**



EMPIRICAL RISK MINIMIZATION

- The risk surface visualizes the empirical risk for all possible parameter values of the parameter vector θ
- Minimizing the empirical risk is often done by numerical optimization

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \mathcal{R}_{\text{emp}}(\theta)$$



Introduction to Machine Learning

ML Basics

Supervised Regression

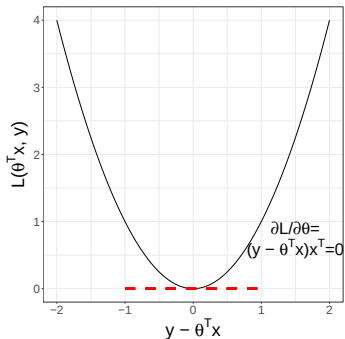
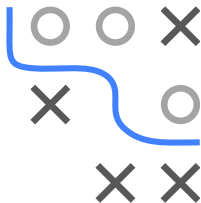
Supervised Classification

Random Forests

Neural Networks

Introduction to Machine Learning

Supervised Regression In a Nutshell



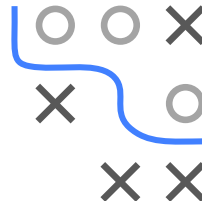
Learning goals

- Understand basic concept of regressors
- Understand difference between L1 and L2 Loss
- Know basic idea of OLS estimator

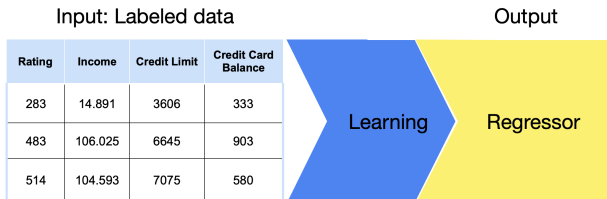
LINEAR REGRESSION TASKS I

- Learn linear combination of features for predicting the target variable
- Find best parameters of the model by training w.r.t. a loss function

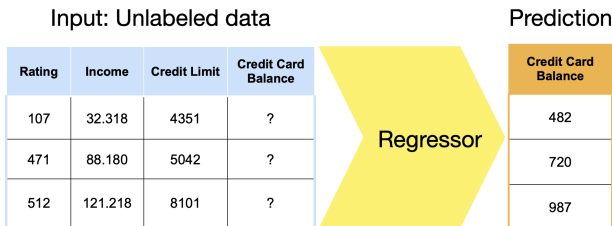
$$\text{CreditBalance} = \theta_0 + \theta_1 \text{Rating} + \theta_2 \text{Income} + \theta_3 \text{CreditLimit}$$



Training

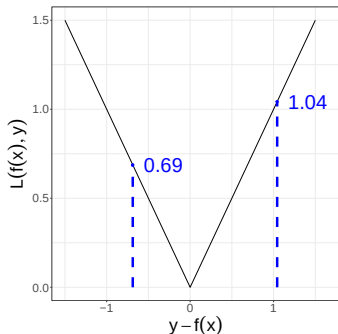
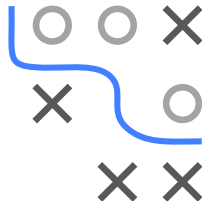


Prediction

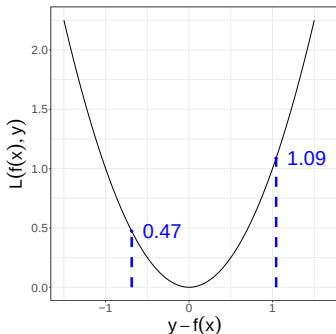


LINEAR MODELS: L1 VS L2 LOSS I

Loss can be characterized as a function of residuals $r = y - f(\mathbf{x})$



- **L1** penalizes the **absolute** value of residuals
- $L(r) = |r|$
- Robust to outliers

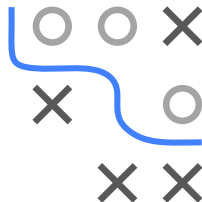


- **L2** penalizes the **quadratic** value of residuals
- $L(r) = r^2$
- Easier to optimize

LINEAR MODELS: L1 VS L2 LOSS I

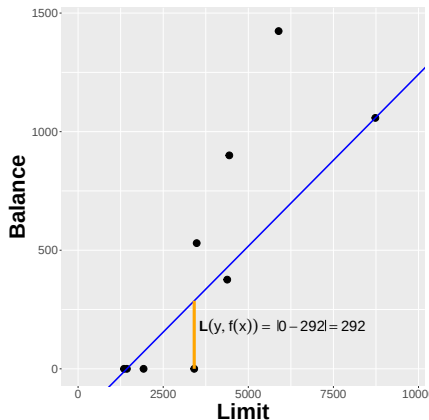
- **L1** Loss is not differentiable in $r = 0$
- Optimal parameters are computed numerically

- **L2** is a smooth function hence it is differentiable everywhere
- Optimal parameters can be computed analytically or numerically

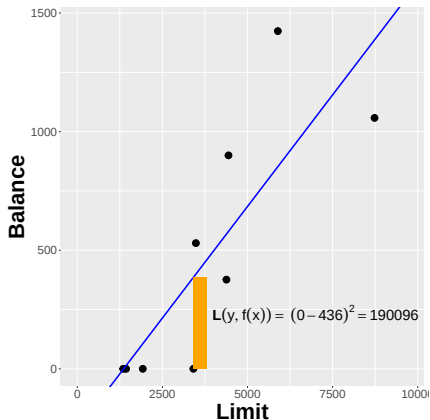


LINEAR MODELS: L1 VS L2 LOSS I

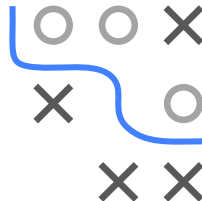
- The parameter values of the best model depend on the loss type



- $\hat{\theta}_{L_1} = 0.14 \rightarrow$ if the Credit Limit increases by 1\$ the Credit Balance increases by 14 Cents



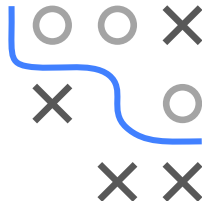
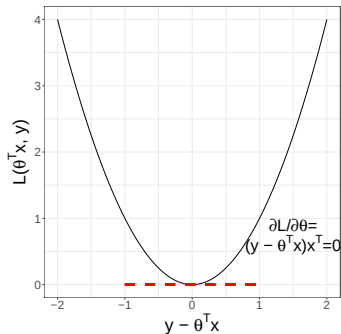
- $\hat{\theta}_{L_2} = 0.19 \rightarrow$ if the Credit Limit increases by 1\$ the Credit Balance increases by 19 Cents



OLS ESTIMATOR I

Ordinary-Least-Squares (OLS) estimator:

- Analytical solution for linear models with L2 loss
- Best parameters can be computed via derivation of the empirical risk
- Solution: $\hat{\theta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$



OLS ESTIMATOR I

Components of **OLS** estimator:

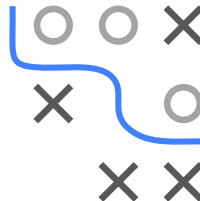
- **X**: Features + extra column for intercept
- **y**: Label vector

 X

Intercept	Rating	Income	Credit Limit
1	283	14.891	3606
1	483	106.025	6645
1	514	104.593	7075

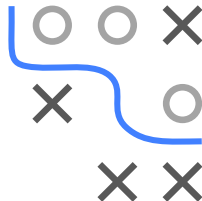
 y

Credit Card Balance
333
903
580

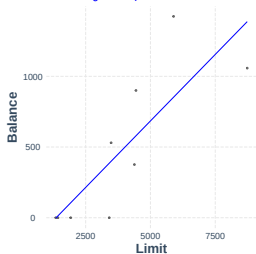


POLYNOMIAL REGRESSION I

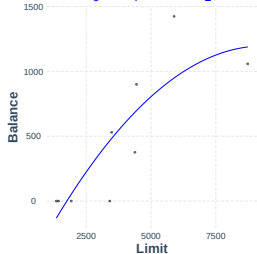
- Adding polynomial terms to the linear combination leads to more flexible regression functions
- Too high degrees can lead to overfitting



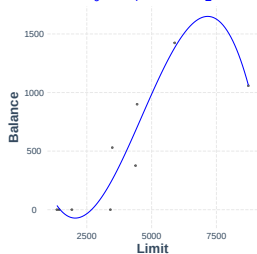
$$\text{Balance} = \theta_0 + \theta_1 \text{Limit}$$



$$\text{Balance} = \theta_0 + \theta_1 \text{Limit} + \theta_2 \text{Limit}^2$$



$$\text{Balance} = \theta_0 + \theta_1 \text{Limit} + \theta_2 \text{Limit}^2 + \theta_3 \text{Limit}^3$$



Introduction to Machine Learning

ML Basics

Supervised Regression

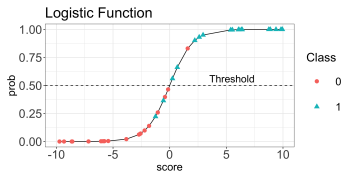
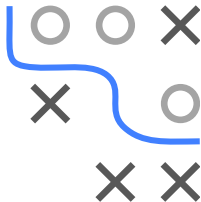
Supervised Classification

Random Forests

Neural Networks

Introduction to Machine Learning

Supervised Classification In a Nutshell

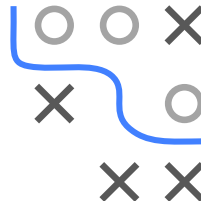


Learning goals

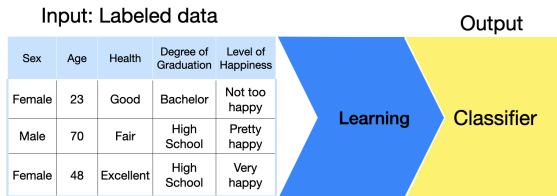
- Understand basic concept of classifiers
- Know concepts of probabilistic and scoring classifier
- Know distinction between discriminant and generative approach
- Understand ideas of logistic regression and Naive Bayes

CLASSIFICATION TASKS I

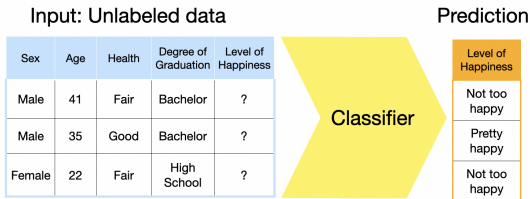
- Learn function that assigns categorical class labels to observations
- Each observation belongs to exactly one class
- The task can contain two (binary) or multiple (multi-class) classes



Training

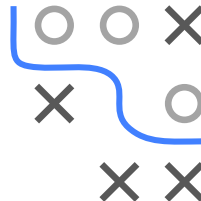


Prediction



BASIC DEFINITIONS I

- For every observation a model outputs the probability (probabilistic classifier) or score (scoring classifier) of each class
- In the multi-class case, the class label is usually assigned by choosing the class with the maximum score or probability
- In the binary case, a class label is assigned by choosing the class whose probability or score exceeds a threshold value c



Input: Unlabeled data

Sex	Age	Health	Degree of Graduation	Level of Happiness
Male	41	Fair	Bachelor	?

Classifier

Class Probabilities

Probability	Level of Happiness
0.4	Not too happy
0.35	Pretty happy
0.25	Very happy

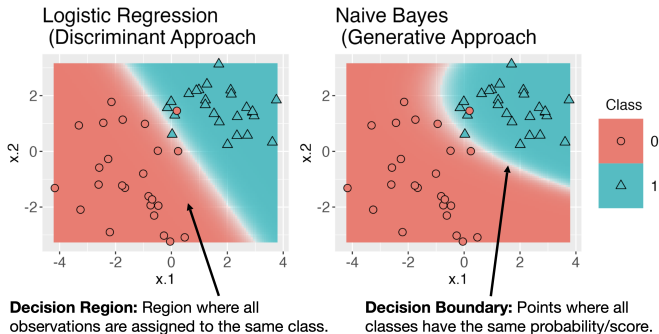
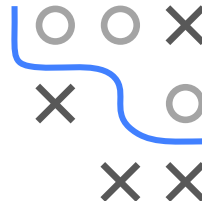
Assigned Label

Level of Happiness
Not too happy

BASIC DEFINITIONS II

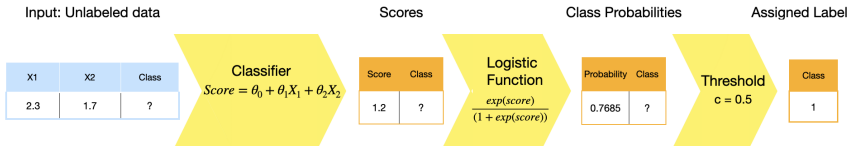
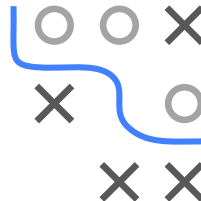
Two fundamental approaches exist to construct a classifier:

- **Discriminant approach** asks “What is the best prediction for the class given these data?” (uses loss functions and empirical risk minimization)
- **Generative approach** asks “Which class tends to have data like these?” (models the feature distributions in each class separately)

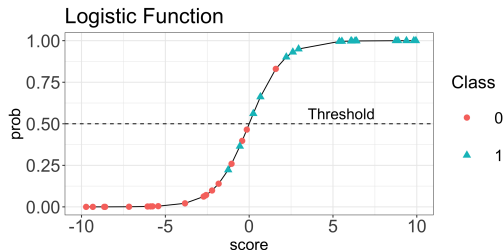


LOGISTIC REGRESSION I

- Logistic regression is a **discriminant approach** for binary classification. It turns scores into probabilities with the logistic function.
- We just need to compute the probability for **one** class (usually class 1).
- If the probability exceeds a threshold value **c** $\Rightarrow$ class 1 is predicted.

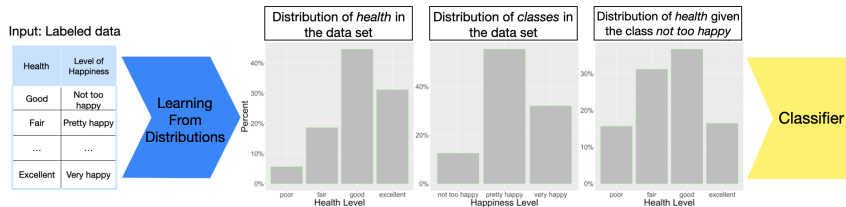
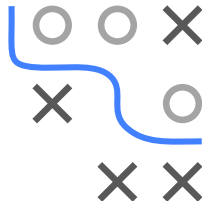


The logistic function puts all scores in order along an s-shaped line.



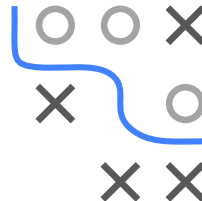
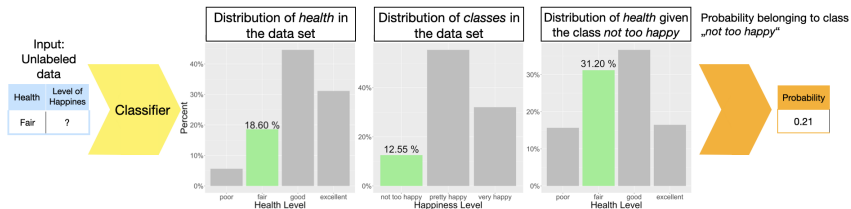
NAIVE BAYES I

- Naive Bayes is a **generative multi-class approach**. It computes the class probability for each class based on the training data.
- It considers the data distribution on three different levels:
 - Marginal distributions $\mathbb{P}(X)$ of each feature (in the entire data set)
 - Marginal distribution $\mathbb{P}(Y)$ of classes (in the entire data set)
 - Conditional distributions $\mathbb{P}(X|Y)$ of each feature in each class

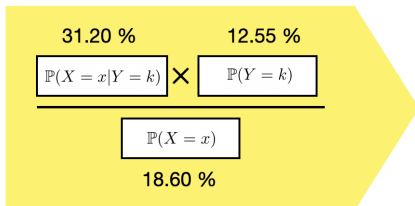


NAIVE BAYES II

- Example: Class probability of “not too happy” given health = “fair”:



Naive Bayes Classifier



Class probability given the data

21.00 %

Introduction to Machine Learning

ML Basics

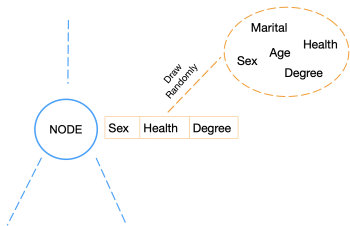
Supervised Regression

Supervised Classification

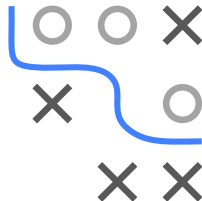
Random Forests

Neural Networks

Introduction to Machine Learning

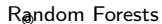


Learning goals



A 3x3 grid with a blue path starting at the top-left cell (0,0) and ending at the bottom-right cell (2,2). The path moves right from (0,0) to (0,1), then down to (1,1), then right to (1,2), and finally down to (2,2). The cells (0,2), (1,0), and (2,0) are marked with 'X', while the other cells are empty.

- ## Training



LEARNING AND PREDICTION WITH RF I

Prediction

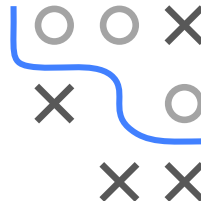
Input: Unlabeled data

Age	Health	Degree of Graduation	Level of Happiness
41	Fair	Bachelor	?
35	Good	Bachelor	?
22	Fair	High School	?

Random
Forest
Model

Prediction

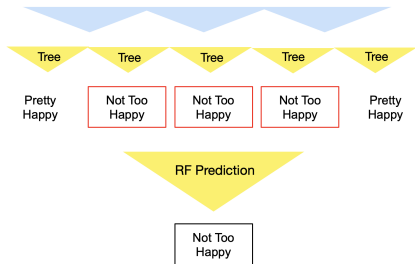
Level of Happiness
Not too happy
Pretty happy
Not too happy



AGGREGATION RULES FOR DIFFERENT TASKS I

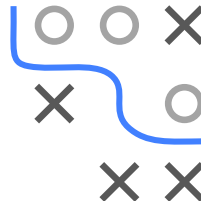
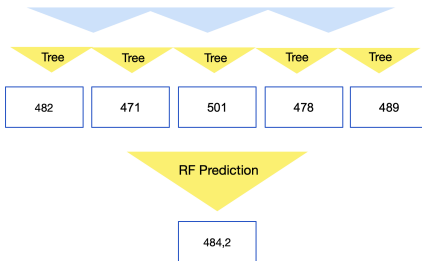
Classification Task - Majority Vote

Age	Health	Degree of Graduation	Level of Happiness
41	Fair	Bachelor	?



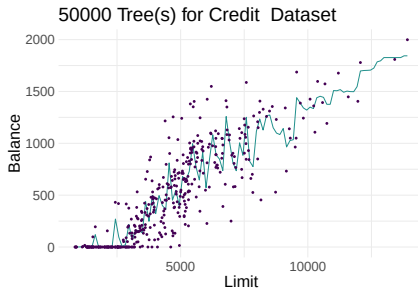
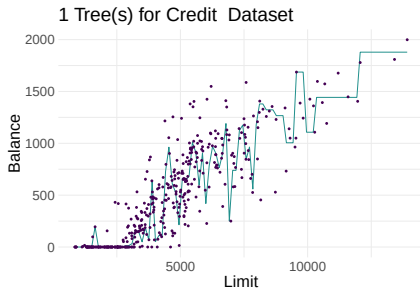
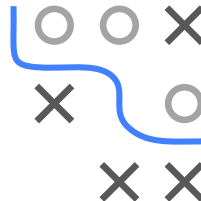
Regression Task - Averaging

Rating	Income	Credit Limit	Credit Card Balance
107	32.318	4351	?



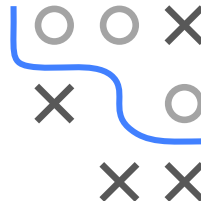
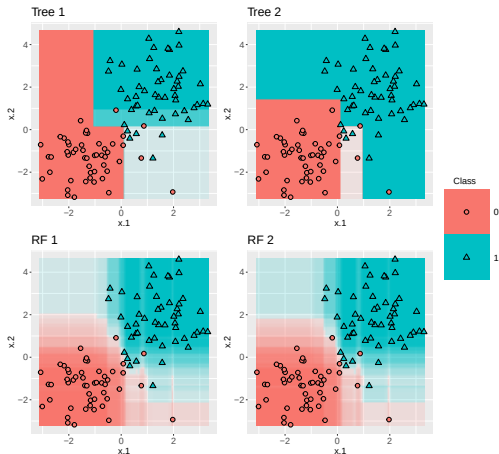
PERFORMANCE OF RF I

- In general: Increasing the ensemble size stabilizes the predictions
 - For regression tasks the stabilization is often not sufficient.



PERFORMANCE OF RF I

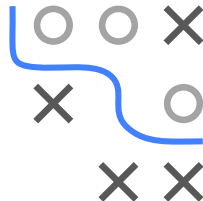
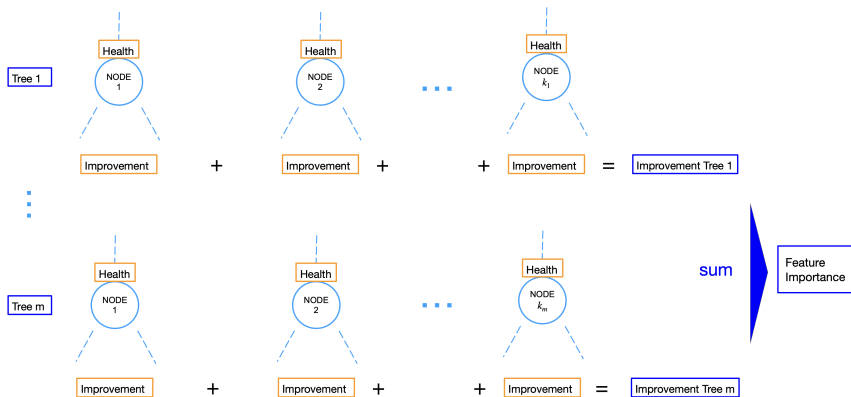
- RF performs well for classification tasks:
 - Two different trees → Quite different decision regions
 - Two different RFs → Similar decision regions



FEATURE IMPORTANCE I

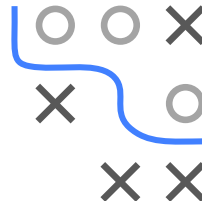
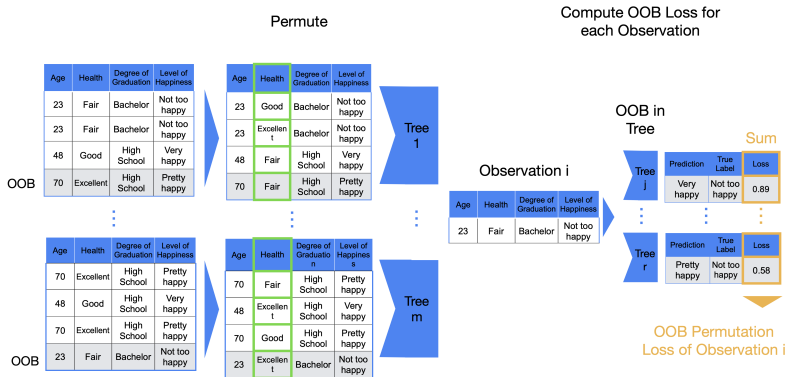
Several options, e.g., measure contribution of feature to model:

- Measure based on improvement in splitting criterion
- E.g. Feature importance of 'Health', search all nodes with 'Health' as splitting variable:

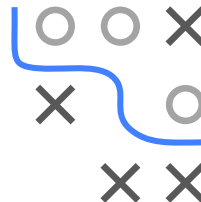
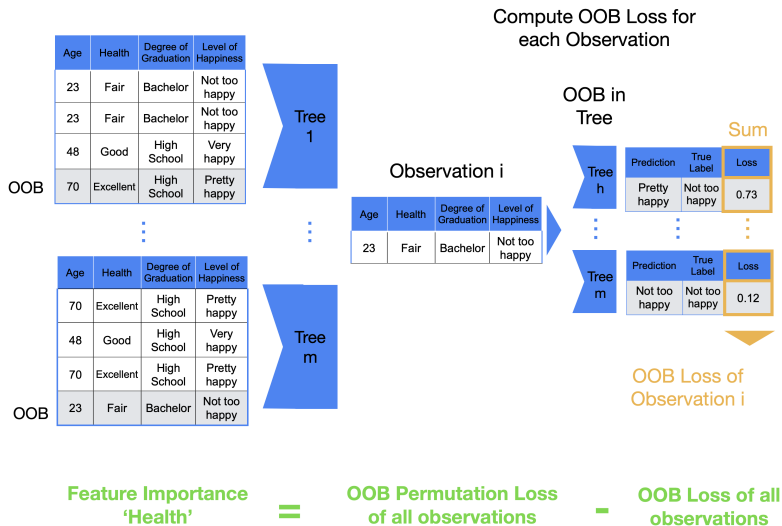


FEATURE IMPORTANCE I

- Measure based on OOB Loss



FEATURE IMPORTANCE I



Introduction to Machine Learning

ML Basics

Supervised Regression

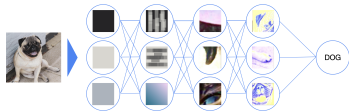
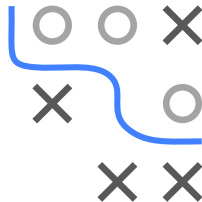
Supervised Classification

Random Forests

Neural Networks

Introduction to Machine Learning

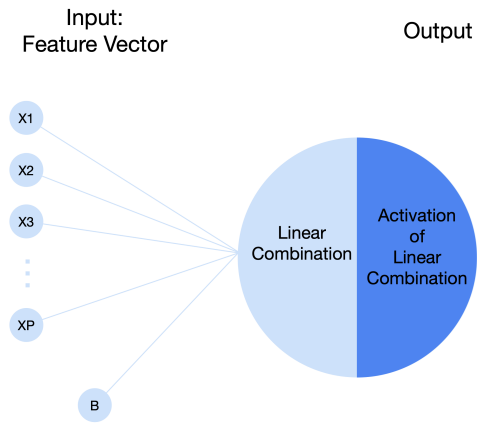
Neural Networks In a Nutshell



Learning goals

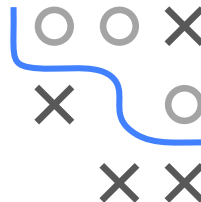
- Know basic computational unit
- Know basic architecture
- Understand Learning in NNs

BASIC COMPUTATIONAL UNIT: PERCEPTRON I



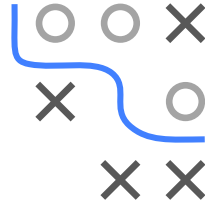
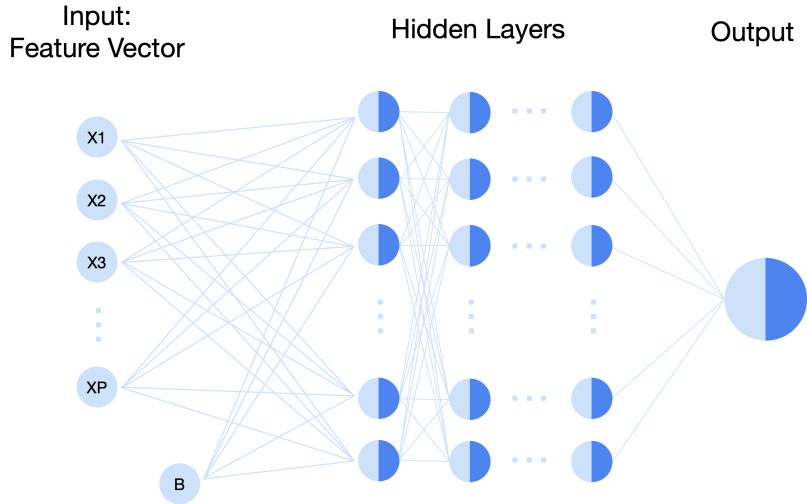
Output differs depending on activation function:

- **Identity function:**
Perceptron represents linear regression
- **Logistic function:**
Perceptron represents logistic regression
- Other activation functions possible



BASIC ARCHITECTURE OF NN I

A neural network is built by combination of multiple perceptrons:



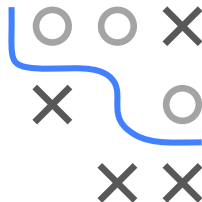
BASIC ARCHITECTURE OF NN I

Hidden Layers:

- Output of hidden units serves as input for units in next layer
- Too many hidden layers or too many units per layer can lead to overfitting

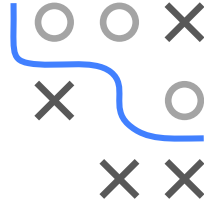
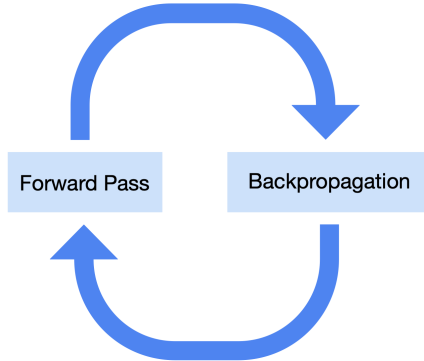
Output Layer:

- Number of output units depend on task
- Different activation functions for output layer and hidden layers possible

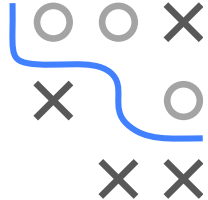
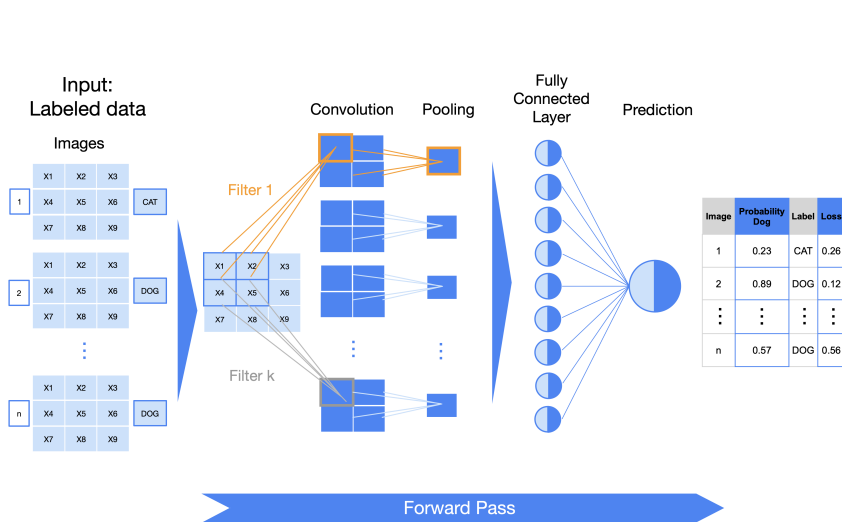


LEARNING - IMAGE CLASSIFICATION TASK I

For each Training Iteration:

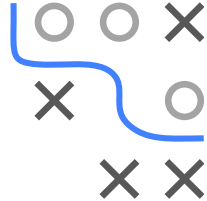
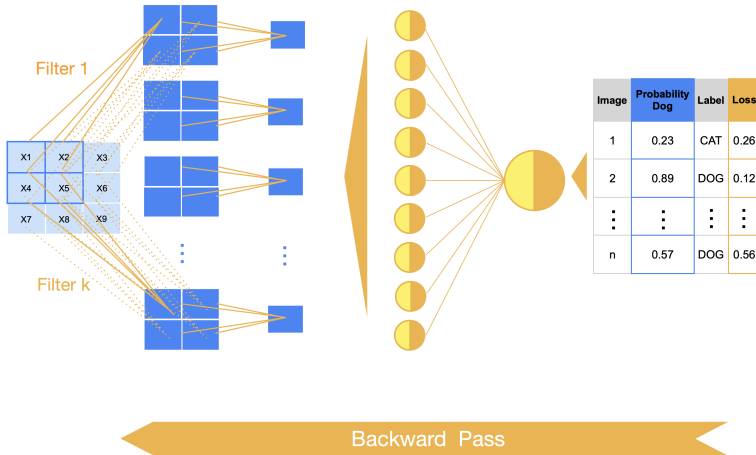


LEARNING - IMAGE CLASSIFICATION TASK I

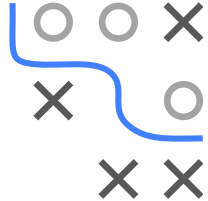
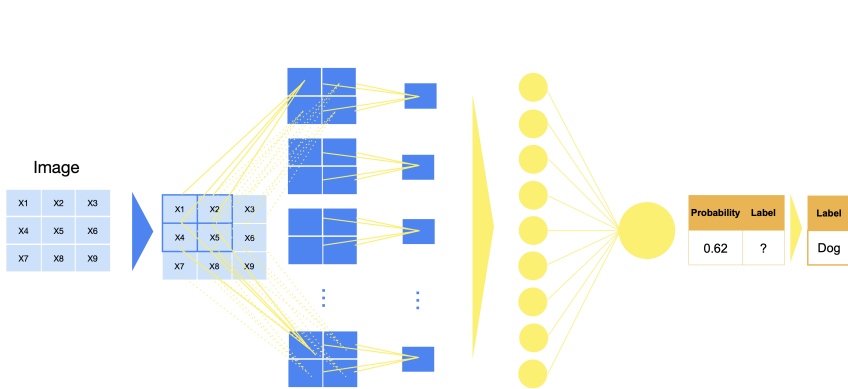


LEARNING - IMAGE CLASSIFICATION TASK I

Compute update of each weight by backpropagation



PREDICTION - IMAGE CLASSIFICATION TASK I



EFFECT OF HIDDEN LAYERS I

- Learn more and more abstract representations
- Each layer adds degree of non-linearity

